

What Does It Mean for the Past to Still Matter?

A Historical-Relevance Audit of Causal Persistence, Information, and Control

Abstract

Something can have happened earlier without remaining present in every scientifically relevant sense. An earlier distinction may still make a controlled causal difference to a later target; information about it may be available now; that information may be retained from the earlier event or reconstructed from present cues; it may change what an agent should do; and it may or may not actually control behavior. Ordinary expressions such as "it still matters" and "the system remembers it" can collapse these relations. This article develops a historical-relevance audit that keeps them separate without proposing new primitives for causation, memory, or decision making. Two constructive cases motivate the synthesis. First, aggregate endpoint control need not identify which temporally indexed intervention remains causally consequential. Second, a cognitive process can possess an objective and recoverable history without thereby possessing an online representation of its position in that history, and available temporal information need not be authoritative for continuation or closure. A matched-present construction then shows how systems can produce the same ordinary observations while differing in external causal persistence, provenance of present historical information, decision authority, and realized use. The methodological consequence is simple: evidence identifying one present relation to history does not identify the others without additional assumptions or interventions. No participant data are reported; the proposed cognitive dissociations remain open empirical questions.

Keywords: historical relevance; causal inference; memory; representation; credit assignment; control; identifiability

1. Introduction: What Does It Mean for the Past to Still Matter?

Something happened before. In what sense is it still here now?

The question sounds singular. Scientifically, it is not. A past event can leave a physical consequence that still changes a present outcome. It can leave a memory after its external consequence has disappeared. A system can reconstruct the correct historical fact from present evidence without preserving the queried trace from the original event. Historical information can be available but irrelevant to the decision currently faced. It can be relevant to the correct decision but fail to control actual behavior. A system can also continue to act on historical information after that information has become obsolete.

Each case can be described informally by saying that the past "still matters" or is "still there." Yet the evidential demands differ. A causal-effect claim requires a controlled comparison on a declared target. An access claim requires evidence that the system can obtain the relevant information. A provenance claim asks why that information is available now. A decision-relevance claim depends on the current objective. A realized-use claim requires showing that changing the information changes the actual policy while competing cues are controlled.

This article develops a common audit for these historically oriented claims. It does not propose a universal theory of memory, causation, or cognition. The component sciences are mature. Markov and partially observable control models formalize when a present information state is sufficient for prediction or control (Kaelbling et al., 1998). Computational mechanics and predictive-state approaches compress histories according to predictive consequences rather than preserving literal records of the past (Littman et al., 2001; Shalizi & Crutchfield, 2001). Causal-abstraction frameworks ask which interventions and causal queries

survive changes in level of description (Beckers & Halpern, 2019; Rubenstein et al., 2017; Xia & Bareinboim, 2025), and recent compositional work makes query sets and semantics explicit (Lorenz & Tull, 2026). Temporal credit assignment treats relations between earlier actions and delayed outcomes as a central problem (Moran et al., 2021; Pignatelli et al., 2024). Source monitoring constrains inferences from remembered content to its origin (Johnson et al., 1993), while causal influence diagrams distinguish information, decisions, utilities, response incentives, and control incentives (Everitt et al., 2021).

The purpose is not to rename these results. It is to connect them around one recurring inferential failure: evidence establishing one present relation to history is often interpreted as though it established another.

Two source problems make the failure especially visible. The first is causal. A temporally factored sequence of interventions can be compressed into one compound input to an endpoint channel. Aggregate endpoint control may remain matched even when a particular earlier intervention persists in one system and reconverges in another. Endpoint-level causal capacity therefore need not identify which earlier distinction still has causal support at the declared endpoint.

The second is representational and functional. A bounded cognitive episode necessarily unfolds through physical time and internal state changes, but process history does not by itself establish an accessible coordinate locating the ongoing episode in elapsed time. A later duration judgment may be reconstructed from event structure or remembered content (Block & Zakay, 1997; Boltz, 1995). Even when precise temporal information is available, availability does not establish that elapsed time should influence the current continuation policy, nor that it actually does. Existing prospective-memory research demonstrates strategic time monitoring and shifts between internal and external timing strategies (Huang et al., 2014; Munaretto et al., 2022; Waldum & Sahakyan, 2013), but the specific access-by-authority dissociation considered here remains an empirical proposal.

Taken together, these problems suggest a broader methodological object. A historical distinction can have different destinations in the present: an external target, a present information state, an objective that defines what should happen next, or an actual controller that determines what does happen next. These routes often agree under ordinary conditions and therefore become easy to collapse.

The central discipline is:

Do not infer one form of historical relevance from evidence that identifies only another.

That claim would be trivial if it remained a plea for terminology. The paper therefore gives it constructive content. It declares the target of each relation, identifies the intervention needed to distinguish neighboring interpretations, gives matched counterexamples, and states when the distinctions should collapse. The contribution is the connection and the resulting identification discipline; stronger formal novelty is secondary.

2. Established Science Already Compresses History

Science routinely replaces histories with present states. In a Markov description, the current state is sufficient for the modeled future conditional on that state. In a partially observable Markov decision process, a belief state summarizes observation-action history for control (Kaelbling et al., 1998). Computational mechanics groups histories that induce the same conditional distribution over futures, while predictive-state representations encode state through action-conditional predictions of future observations (Littman et al., 2001; Shalizi & Crutchfield, 2001).

Causal abstraction adds a different requirement: a coarser description must preserve the intervention distinctions and causal queries intended at the higher level. Exact transformations and later abstraction frameworks make intervention preservation central (Beckers & Halpern, 2019; Rubenstein et al., 2017). Projected causal abstractions extend the program to lossy representations in which distinct low-level interventions can map to a shared high-level description while selected observational, interventional, and counterfactual queries remain translatable (Xia & Bareinboim, 2025). Lorenz and Tull (2026) formulate abstraction relative to declared query sets and semantics.

These frameworks already establish the broad principle that a representation can be sufficient for one purpose while discarding distinctions needed for another. That principle is inherited science, not the novelty claim here. The narrower issue is what follows when several queries concern the same historical distinction but target different present relations. A state may be sufficient to predict the next observation without preserving which earlier intervention is still causally supported. A present pattern may permit an external decoder to recover a historical feature without showing that the system can access or use it (Ritchie et al., 2019). A correct report may be supported by retention or by reconstruction. A historical feature may matter to the optimal policy under one objective and be irrelevant under another.

Historical relevance is therefore not treated here as an intrinsic property of a past event. It is a relation between a historical distinction and a declared present target, information pathway, objective, or policy.

3. A Historical-Relevance Audit

Let H denote a scientifically chosen historical distinction: an earlier action, cue, perturbation, environmental state, temporal coordinate, or another past difference for which an intervention or contrast is meaningful. Let Z_t denote a declared present or endpoint target. Let M_t denote present information usable by the system and concerning H or a property of H . Let D_t denote a current decision, continuation choice, stopping action, or policy output. Let U denote the present objective or criterion under which actions can be evaluated.

Four questions survive the audit. They are not claimed as new primitives.

3.1 External causal persistence

Would changing H , under the relevant controlled comparison, still change Z_t ?

For historical values h and h' and comparison variables W fixed at w , causal persistence to Z_t exists when

$$P(Z_t \mid \text{do}(H = h), \text{do}(W = w)) \neq P(Z_t \mid \text{do}(H = h'), \text{do}(W = w))$$

for at least one relevant w . The relation is explicitly target-relative. H can have reconverged with respect to one target while remaining causally ancestral to another. Saying that the past is simply "causally gone" is incomplete unless the target and intervention set are named.

3.2 Present historical information and provenance

Does the system currently possess or can it obtain usable information about H , and what route made that information available?

Availability does not determine provenance. A current representation M_t can be a causal descendant of H , or the same historical answer can be reconstructed from current cues correlated with H . Source-monitoring research already treats judgments about origin as inferential and fallible (Johnson et al., 1993). The narrower causal audit asks whether changing H while alternative reconstructive routes are held fixed changes M_t . If it

does, H retains controlled support to the present representation under that comparison. If it does not, information about H may still be accurate, but another route supplies the accuracy.

This matters whenever a correct report is used as evidence for an earlier online representation. A report can establish recoverability at report time without identifying how or when the information was represented.

3.3 Decision authority

Would historical information change what the system should do under the declared objective?

Let $D^*(m, s; U)$ denote an optimal decision given historical information m , other present information s , and objective U . Historical information has authority where $D^*(m, s; U)$ differs from $D^*(m', s; U)$ for relevant m and m' . This is objective-relative and closely related to established value-of-information and response-incentive questions (Everitt et al., 2021). The term is retained because the temporal case requires a compact distinction between "time is available" and "time belongs in the continuation policy." It is not a new decision-theoretic primitive.

3.4 Realized use

Does the actual policy depend on the historical information now?

A direct use test changes the present historical signal while experimentally holding a declared set of competing present information fixed. Let S_t denote that competing present information/cue set. For relevant m, m' , and s , realized use is supported when

$$P(D_t \mid \text{do}(M_t = m), \text{do}(S_t = s)) \neq P(D_t \mid \text{do}(M_t = m'), \text{do}(S_t = s)).$$

Even this does not establish that the resulting action is the action warranted by the corresponding history. Zhang et al. (2026) make this distinction explicit in history-conditioned robot policies by crossing lawful histories at an identical present and separating memory sensitivity, history-warranted choice, physical value, and reliability. The contrast is intervention- and architecture-relative: S_t must name a competing cue set that can be meaningfully controlled without changing the historical relation being tested.

Predictive relevance remains important but is not promoted to a fifth core relation. Markov sufficiency, computational mechanics, predictive-state representations, and information theory already provide mature ways to ask whether history improves prediction of a declared future target. Prediction therefore functions here as an ownership boundary and comparison class.

4. Constructive Causal Case: Endpoint Control Does Not Identify Temporal Support

Consider a temporally factored intervention history $U_T = (A_0, \dots, A_{(T-1)})$ and an endpoint kernel $K_T(e \mid \text{do}(a_0, \dots, a_{(T-1)}))$. Treating the complete history as one compound channel input permits endpoint-control quantities such as channel capacity to summarize how much distinguishable influence the compound history can exert on the endpoint. The same summary need not preserve which temporal coordinate carries that influence.

To see the identification problem, let A_0, A_1 , and A_2 be binary. Define shared early states

$$E_1 = A_0$$

$$E_2 = (A_0, A_1).$$

Define two endpoints:

$$\text{System P: } E_3^P = (A_0, A_2)$$

System M: $E_3^M = (A_1, A_2)$.

Both endpoints have four reachable states and therefore the same deterministic endpoint capacity of 2 bits. Swapping A_0 and A_1 in the compound input maps one endpoint channel to the other, so the channels are isomorphic when complete histories are treated only as atomic input labels.

The causal status of A_0 nevertheless differs maximally. In P, changing A_0 while holding A_1 and A_2 fixed always changes the endpoint. In M, changing A_0 under the same controlled comparison never changes the endpoint. The A_0 distinction exists earlier in the trajectory but has reconverged before E_3 . The difference also survives a shared stochastic post-channel whenever the A_0 contrast remains detectable in P.

The bounded result is:

Endpoint causal capacity does not, in general, identify its temporal factorization.

This does not identify a psychological mechanism or representation. It makes one sense of "still matters" precise: a historical distinction remains causally relevant to a declared target when the target still changes under a controlled intervention on that distinction. Temporal credit-assignment research already owns the broader problem of linking earlier actions to delayed outcomes (Moran et al., 2021; Pignatelli et al., 2024), and causal abstraction already owns the generic requirement that an abstraction preserve the intervention distinctions needed by the query. The role of the construction here is to provide a concrete causal lane for the larger historical audit.

5. Constructive Representational Case: Process History Is Not Temporal Self-Location

A bounded cognitive episode beginning at t_0 and closing at t_1 necessarily has objective duration $T = t_1 - t_0$. It also has a process history: a trajectory through evidence states, candidate interpretations, confidence, remembered subevents, or other task-relevant variables. Neither fact establishes that the ongoing episode has a usable representation of where it is in elapsed time.

This is an identification problem, not a claim that cognition lacks timing mechanisms. A participant may later estimate duration accurately because elapsed time was monitored online, because event structure was remembered, because task regularities support reconstruction, because a clock was queried, or through some combination. Prospective and retrospective timing research makes reconstruction a default competitor rather than an exotic alternative (Block & Zakay, 1997; Boltz, 1995). A later report therefore identifies recoverability at report time more directly than it identifies an online temporal coordinate.

The temporal case also separates access from authority. Let access mean that precise elapsed-time information can be obtained during the episode, for example through an on-demand query. Let temporal authority mean that elapsed time is relevant to the optimal continuation or closure policy. A clock can be available when elapsed time should not change the decision. Conversely, elapsed time can matter to the task when precise external time is unavailable, requiring internal or indirect cues.

Existing time-based prospective-memory research already demonstrates strategic clock monitoring, effects of consequences on monitoring, memory contributions to prospective timing, and shifts between more internal and external forms of control (Huang et al., 2014; Munaretto et al., 2022; Waldum & Sahakyan, 2013). The narrower empirical proposal is to cross precise temporal access with whether elapsed time has authority in a self-terminated deliberative episode while treating event-history reconstruction as a competing route. No participant dataset currently establishes that dissociation.

The relevance to the present synthesis is straightforward. Process history does not establish present historical information in the queried form; access to such information does not establish its decision authority; and authority does not establish realized use.

6. Historical Information Does Not Identify Its Provenance

The connection between the two source cases becomes sharper when the target is not an external endpoint but a present representation of history.

Let H be an earlier binary distinction and C a later/current cue. Consider two systems that give the same correct report under ordinary conditions.

Retained-history system:

$H \rightarrow M \rightarrow \text{report}$.

Reconstructed-history system:

$L \rightarrow H$ and $L \rightarrow C \rightarrow M \rightarrow \text{report}$,

with no $H \rightarrow M$ path in the queried representation. When ordinary conditions satisfy $H = C$, both systems can report H perfectly. Accuracy alone cannot identify whether M is a surviving causal descendant of H or a reconstruction from a current cue.

The discriminating intervention is to vary H while holding the reconstructive route fixed, or otherwise decorrelate H from C . If M follows H , evidence for retained historical provenance strengthens. If M follows C , the correct ordinary report was reconstructive relative to the tested route. This is the same intervention-relative logic used to ask whether an earlier coordinate still has causal support at an external endpoint, now applied to the present historical representation itself.

The point does not claim that reconstruction is defective. A reconstruction may be highly accurate and functionally useful. The claim is only that "the system knows what happened" does not by itself identify why the system can answer correctly now.

7. One Matched-Present Construction for the Whole Problem

A single construction can place the causal, informational, authority, and use questions in one system.

Let an earlier historical distinction H and a later/current cue C normally be generated by the same source L , so that ordinary operation gives $H = C = L$. Add a present external target Z , a present historical representation M , and a decision D . Define four structural switches:

p : Z follows H when $p = 1$ and C when $p = 0$.

r : M follows H when $r = 1$ and C when $r = 0$.

u : D follows M when $u = 1$ and C when $u = 0$.

a : reward is maximal for $D = H$ when $a = 1$ and for $D = C$ when $a = 0$.

Under the ordinary regime $H = C = L$, all $2^4 = 16$ assignments of (p, r, u, a) can yield $Z = M = D = L$ and maximal reward. Correct endpoint, correct report, correct action, and successful performance are therefore compatible with structurally different relationships to the same past.

The structures separate only when the relevant redundancy is broken. Setting $H \neq C$ identifies whether Z or M follows the historical or current coordinate, thereby auditing external persistence and information

provenance. Holding the world fixed while setting $M \neq C$ identifies whether actual decisions follow the historical representation or the competing present cue. Creating H-C conflict under different objectives identifies which information should govern the decision.

This finite witness is not offered as a new general theorem of cognition. Its value is diagnostic. It explains why several true ordinary observations can coexist while leaving the relations among them unidentified. The same present can be compatible with different answers to "what from the past is still here?"

8. What Evidence Identifies Which Claim?

The construction yields a practical audit. Evidence should be interpreted at the relation it identifies rather than promoted automatically to a neighboring claim.

Claim	Immediate target	Identifying evidence/intervention	What remains unlabeled
External causal persistence	$H \rightarrow Z$	Controlled intervention on H with relevant alternatives fixed	Access, representation, authority, actual use
Historical information / provenance	$H \rightarrow M$ or reconstruction $\rightarrow M$	Assay access; decorrelate H from reconstructive cues	Online maintenance, authority, actual use
Decision authority	U-relative optimal policy	Change objective or historical information and recompute optimal action	Actual access or realized use
Realized use	$M \rightarrow D$	Intervene on M while experimentally fixing the declared competing cue set S	Warranted choice, value, consciousness

Several familiar observations therefore have bounded interpretations. Endpoint success can establish competence or endpoint influence without identifying which earlier coordinate remains causally supported. A correct historical report establishes recoverability more directly than continuous online maintenance or provenance. External decodability identifies information available to the decoder, not necessarily representation or use by the system (Ritchie et al., 2019). Optimal-policy relevance establishes authority under the declared objective but not actual access or implementation. Behavioral sensitivity to a memory channel establishes causal dependence on that channel, not that the resulting action is the one warranted by the matching history (Zhang et al., 2026).

9. Collapse Conditions

The audit is useful only if its distinctions are allowed to disappear.

First, if a present state is sufficient for every declared predictive, causal, representational, and control query under the relevant interventions, no additional historical variable is scientifically needed for those queries. History can remain genealogically true without functioning as an independent present explanatory variable.

Second, if the equivalence relation preserving prediction also preserves every causal-support query of interest, predictive and causal descriptions need not be separated in that domain. Likewise, if retained and

reconstructed information produce the same action under every admissible intervention and provenance is not itself a scientific target, provenance can be ignored.

Third, if representation and policy use cannot be varied separately in the selected system, the experiment cannot identify them separately. The correct conclusion is local: the chosen architecture does not support the distinction. It does not follow that access and use are universally identical.

Finally, temporal factorization matters only when the temporal coordinates have scientific meaning. If a sequence has no interpretable earlier and later intervention roles, preserving factorization is arbitrary bookkeeping. Authority can also be trivial by construction when every admissible objective either always depends on H or never does. The framework should collapse rather than manufacture distinctions in such cases.

10. Consequences for Psychological Inference and Experimental Design

The audit changes experimental design by locating interventions at the relation being claimed.

For external persistence, choose a historical variable H and target Z, then create conditions in which H persists or reconverges before Z while aggregate endpoint opportunity is matched as closely as possible. The critical outcome is the controlled contrast on Z under intervention on H, not an overall success score.

For historical information and provenance, first assay whether the system can obtain the relevant historical property. Then decorrelate the original history from candidate reconstructive cues. In temporal cognition, this means that an unexpected retrospective duration judgment is secondary evidence: event count, remembered content, and elapsed time should be partially decorrelated if online temporal self-location is the target.

For authority, manipulate the objective so that the same historical information changes the optimal policy in one condition but not another. This keeps "should use" separate from "can access." For realized use, intervene on the candidate historical representation while experimentally fixing the declared competing cue set S and measure the actual policy. The Counterfactual Memory Audit provides a close applied precedent for this fixed-present logic (Zhang et al., 2026).

These modules can be combined without pretending they share one target. A cognitive study might first manipulate whether an earlier contribution remains consequential to a later external state, independently manipulate access to a precise historical cue, and manipulate whether the current objective makes that cue relevant to continuation. Direct perturbation of the present historical signal can then test use. The resulting design would ask four different questions rather than labeling them as one latent quantity of "relevance."

The temporal source case supplies a concrete downstream example. During a sequential self-terminated deliberation, a precise elapsed-time query can be made available or unavailable, and the payoff can independently make elapsed time relevant or irrelevant to the optimal continuation policy. Event timing can be jittered so that elapsed duration is not perfectly indexed by evidence count. Authority-dependent query behavior together with closure effects would support strategic recruitment of precise temporal information; retrospective duration would remain secondary because it can be reconstructed. This Access $\times$ Authority proposal is OPEN until tested with participant data.

11. Ownership Boundaries and Closest Prior Art

The audit deliberately inherits rather than competes with several mature frameworks.

State-sufficiency, causal-state, and predictive-state approaches already show that history can be compressed relative to a target (Kaelbling et al., 1998; Littman et al., 2001; Shalizi & Crutchfield, 2001). Causal-

abstraction work already shows that adequacy depends on preserving the interventions and queries that matter at the selected level (Beckers & Halpern, 2019; Rubenstein et al., 2017; Xia & Bareinboim, 2025), and Lorenz and Tull (2026) make query-relative semantics especially explicit. Consequently, "sufficiency depends on the question" is not a novelty claim of this paper.

Murray (2026) provides a particularly close matched-present/history-conditioned state-sufficiency precedent in biology: after the candidate present is matched, prior history is retained only when it changes a declared future response law. This directly precedents the matched-present predictive-sufficiency lane of the present audit, but it does not by itself identify external causal persistence, provenance of present historical information, objective-relative authority, or realized use.

Memory and representation research own important neighboring distinctions. Source-monitoring theory treats origin judgments as inferential rather than transparent readouts of stored records (Johnson et al., 1993). Decoding work warns that information recoverable by an external classifier need not establish the represented or functionally used content of a neural state (Ritchie et al., 2019). Timing research already shows memory contributions to prospective timing and strategic external monitoring (Huang et al., 2014; Munaretto et al., 2022; Waldum & Sahakyan, 2013). Causal influence diagrams already distinguish information, decisions, value, response incentives, and control incentives (Everitt et al., 2021).

Two recent neighbors are especially close. Zhang et al. (2026) directly precedent the realized-use layer by crossing histories at an identical present and distinguishing memory sensitivity from warranted action and value. Golovko (2026) distinguishes persistence of causal consequences from persistence of evidence sufficient to identify their origins in a different epistemic domain. These works narrow the ownership claim appropriately.

What remains is connective and methodological: one historical distinction is followed simultaneously toward an external causal target, a present information state and its provenance, an objective-relative decision, and an actual policy. The matched-present construction shows why ordinary evidence can leave those routes unresolved, and the audit assigns different interventions to the different claims. Exact prior-art ownership of that full synthesis remains open; the paper does not claim that every component or the generic mathematics is new.

12. Limitations and Open Questions

The framework has several limits. It is supported by constructive cases and neighboring literatures, not by a broad survey showing that researchers routinely commit every prohibited inference. It is therefore an identification framework and design discipline, not a prevalence claim about scientific error.

System boundaries and targets matter. H can have reconverged with respect to an external Z while remaining a causal ancestor of a memory M. Provenance can also be much richer than the binary retained-versus-reconstructed witness: real systems can combine partial traces, current cues, semantic knowledge, generative models, and external records. The witness establishes non-identifiability, not a complete theory of memory formation.

Actual use may be distributed across redundant pathways. Failure to perturb one candidate M does not prove that history is unused if another representation carries equivalent information. Authority is likewise model-relative: what the system "should" do depends on U, which can be plural or changing in natural cognition.

Most importantly, the temporal cognitive case remains empirically open. Existing timing and prospective-memory results support strategic monitoring, but the specific Access $\times$ Authority crossing in self-terminated

deliberation has not been tested as proposed. The synthesis cannot treat that dissociation as established evidence.

Finally, the literature boundary remains revisable. Query-relative abstraction, counterfactual memory-use auditing, and causal-versus-epistemic provenance now approach important pieces of the framework. If future work supplies the same cross-target synthesis more completely, the ownership claim should be reduced rather than protected.

13. Conclusion

Something can happen earlier and remain present in more than one scientifically identifiable sense. Its consequences can remain because changing the earlier distinction would still change a declared target. Information about it can remain because the system can recover a relevant historical property. The provenance of that information can trace to the original event, or the same answer can be reconstructed from another present source. The information can matter to what should happen next because the objective makes it decision-relevant. And it can matter to what actually happens next because the current policy uses it.

These relations often align. That is why ordinary language collapses them. The matched-present construction shows that they need not align: different historical structures can produce the same endpoint, report, action, and reward until a relation-specific intervention breaks the redundancy.

The general machinery is largely established. Markov and predictive states explain history compression; causal abstraction explains query-sensitive preservation; credit assignment treats delayed influence; source monitoring constrains historical-report inferences; causal influence diagrams formalize information and decision relevance; timing research studies strategic temporal access; and counterfactual memory auditing tests whether memory guides warranted action.

The contribution is to connect those pieces around one question: **Something happened before. In what scientifically identifiable sense is it still here now?** A useful answer begins by naming the target, then asks what information is presently available, where that information came from, whether the current objective gives it authority, and whether the actual controller uses it. Evidence for one relation is not evidence for the next unless the bridge has itself been identified.

Declarations

Data availability

No new empirical data were created or analyzed in this theoretical study.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT for research assistance, conceptual formalization, literature discovery and organization, adversarial critique and stress testing, drafting and editing assistance, notation refinement, and submission preparation. The author originated and directed the research questions, reviewed and adjudicated model outputs, independently checked cited sources and factual claims, edited the manuscript as needed, and takes full responsibility for the content of the article. ChatGPT is not an author, and no participant dataset was generated by AI.

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